



NEWS HEADLINE ANALYZER



Presented By

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Date

June 12, 2026



Domain

Python

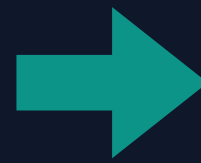
Meet the Intern



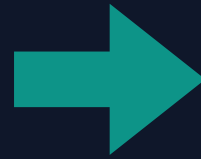
Joined: 8th June, 2026



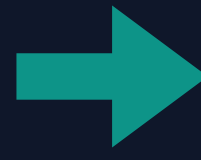
Journey So Far



Name: Gayatri Semwal



Role : Data Science Intern



Objective: To master Python and solve real world problems through hand on projects.



Organization: TechRBM

Topics Covered So Far - My Learnings

Python Basics



Variables, data types, operators, input/output, conditionals & loops

Data Structures



Lists, tuples, sets, dictionaries — operations and use cases

News Headline Analyzer



Applied all concepts in a full project — categorization, search, frequency analysis



What is News Headline Analyzer?



A Python-based command-line tool that classifies, searches, and analyzes a list of news headlines using dictionaries, sets, and string operations — with no external libraries.



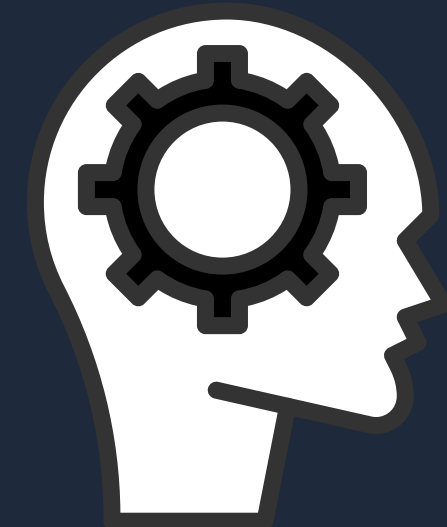
Role

Classifies headlines into 9 categories: Technology, Health, Sports, Finance, and more using keyword matching.



Purpose

Helps users quickly identify the most frequent topics, similar stories, and most covered news categories.



Core Logic

Uses sets for category keywords, dictionaries for frequency counting, and nested loops for pairwise comparison.

Code Snippet



news_headline_analyzer.py

```
categories = { 'Technology':{'tech','ai','python'}, ... }
```

```
for headline in headlines:
```

```
    words = headline.lower().split()
```

```
    for category, keywords in categories.items():
```

```
        for word in words:
```

```
            if word in keywords:
```

```
                headline_category[headline] = category
```

```
                break
```

```
most_covered = max(category_count, key=category_count.get)
```


Program Output



Terminal Output

NEWS HEADLINE ANALYZER

- 1. Show all categories
- 2. Top 5 frequent words
- 3. Similar headlines
- 4. Search by keyword
- 5. Categorise my headline
- 6. Most covered topic
- 7. Exit

Enter your choice: 6

🏆 Most covered: Technology
(4 headlines)

14

Headlines

Total analyzed

9

Categories

Classified topics

Tech

Top Topic

4 headlines

Future Scope

01



Live News Integration

API Integration

Connect to NewsAPI to fetch real headlines instead of a fixed list
– making the analyzer dynamic and always up to date.

02

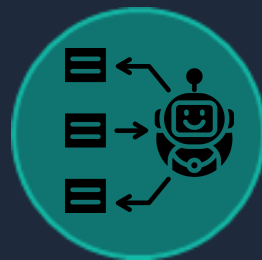


Visual Charts

Data Visualization

Add a Matplotlib bar graph showing category distribution – turning raw numbers into easy-to-read visuals for better insights.

03



Smarter Categorization

NLP / AI

Expand keyword sets or use basic NLP techniques to improve classification accuracy beyond simple keyword matching.



Thank You!

News Headline Analyzer — Day 6 Internship Project

Open to questions & feedback!